

Effects of Cerebral Palsy on Cytoskeleton Rearrangement and E. Coli Susceptibility

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Introduction

The non-progressive neurological disorder cerebral palsy (CP) results from abnormal brain development or injury during early life, leading to persistent impairments in movement, posture, and muscle coordination. Common symptoms include spasticity, involuntary movements, and delayed motor development, though severity varies [1]. CP affects approximately 2–3 per 1,000 live births, with prevalence staying constant in recent years [2]. While the brain injury itself does not progress, functional limitations can worsen over time, often impacting independence. The heterogeneous etiology and wide clinical spectrum complicate diagnosis and management, and curative treatments remain unavailable [3].

Methods

I took samples from the Gene Expression Omnibus (GEO, accession number: GSE31243; accessed March 2025, from a study done by Smith et. al), which contained expression profiling by microarray data from gracilis and semitendinosus muscle samples from CP (n=10) and typically developing control (n=10) children under 18. I used GEO2R to analyze the differential expressed genes between CP gracilis muscle samples and typically developing gracilis muscle samples. Enrichment testing was done using STRING-db to find biological and functional pathways enriched for the significant genes.

Results

I found that there were multiple genes in the cytoskeletal rearrangement pathways of both small and large intestinal epithelial cells that were differentially expressed in CP patients (count in network = 7 of 55, false discovery rate = 0.0034). In particular, WASL was upregulated in CP patients compared to the controls (adjusted p-value = 0.00003, log2FoldChange = 1.856), while TUBA1A (adj-p = 0.003, log2FC = -0.979), TUBA1B (adj-p = 0.004, log2FC = -0.572), TUBB6 (adj-p = 0.004, log2FC = -1.072), YWHAQ (adj-p = 0.002, log2FC = -0.663), YWHAZ (adj-p = 0.005, log2FC = -0.521), and CD14 (adj-p = 0.003, log2FC = -1.026) had lowered levels.

Discussion

The downregulation of CD14 causes reduced activation of the receptor signalling pathway, meaning that inflammation and general immune response is reduced. In addition, the lowered levels of genes in the 14-3-3 family (YWHAQ, YWHAZ) could prevent alterations in the intermediate filament network, which are known to contribute to cell signalling cascades [4]. Meanwhile, the reduced presence of tubulin (TUBA1A, TUBA1B, TUBB6) proteins could result in lowered stress fiber formation, which can reduce a cell's ability to heal itself [5]. The upregulation of WASL could also be damaging the cell by promoting the formation of pedestals, which bacteria like E. coli can use to enhance their ability to infect other cells [6]. Taken together, my results indicate that cerebral palsy may be correlated with extreme escherichia coli susceptibility, by reducing epithelial cells' abilities to communicate and defend themselves against invading bacteria.

While the link between cerebral palsy and E. coli has been understudied, further research should explore this link in an experimental setting to develop infection prevention methods. Further research should also attempt to address the limitations in this study, namely the small sample size (n=10 for both groups) and the difference in age ranges between the patient (12.5) and control (4.6) groups.

Keywords: Cerebral palsy, E. coli, gene expression analyses

References:

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